

Gauss-Codazzi Relation 笔记

这个笔记一开始是为了计算2004.00003和0810.1790中的3+1形式做的，但是后来发现文章里的结果似乎有些问题，目前我也还没把正确的结果搞出来

虽然写的是3+1分解，但实际上过程中会保证得到的结果对高维引力的d+1分解也正确，只是习惯上称呼为3+1分解

考虑3+1分解

$$t^a = Nn^a + N^a$$

外曲率的定义

$$K_{ab} := h_a^c h_b^d \nabla_c n_d = h_a^c \nabla_c n_b$$

黎曼曲率张量的定义（约定）是

$$R_{abcd}n^d = R_{abc}{}^d n_d = (\nabla_a \nabla_b - \nabla_b \nabla_a)n_c$$

空间张量场 $T^{a_1 \dots a_k}_{b_1 \dots b_k}$ 的空间导数定义为

$$D_c T^{a_1 \dots a_k}_{b_1 \dots b_k} := h^{a_1}_{d_1} \dots h^{a_k}_{d_k} h^{e_1}_{b_1} \dots h^{e_k}_{b_k} \nabla_c T^{d_1 \dots d_k}_{e_1 \dots e_k}$$

Sec1 Gauss Relation

从这个定义出发可以得到

$$D_a D_b \omega_c = h_a^e h_b^f h_c^g \nabla_e \nabla_f \omega_g - K_{ac} K_b^m \omega_m + K_{ab} h_c^g n^m \nabla_m \omega_g$$

计算过程

$$\begin{aligned} D_a D_b \omega_c &= D_a [h_b^f h_c^g \nabla_f \omega_g] \\ &= h_a^i h_b^j h_c^k \nabla_i [h_j^f h_k^g \nabla_f \omega_g] \\ &= h_a^i h_b^j h_c^k h_j^f h_k^g \nabla_i \nabla_f \omega_g + h_a^i h_b^j h_c^k h_j^f (\nabla_i h_k^g) \nabla_f \omega_g \\ &\quad + h_a^i h_b^j h_c^k h_k^g (\nabla_i h_j^f) \nabla_f \omega_g \\ &= h_a^e h_b^f h_c^g \nabla_e \nabla_f \omega_g + h_b^j h_c^k n^g K_{ak} \nabla_f \omega_g + h_b^j h_c^g n^f K_{aj} \nabla_f \omega_g \\ &= h_a^e h_b^f h_c^g \nabla_e \nabla_f \omega_g - h_b^f h_c^k K_{ak} \omega_g \nabla_f n^g + K_{ab} h_c^g n^f \nabla_f \omega_g \\ &= h_a^e h_b^f h_c^g \nabla_e \nabla_f \omega_g - K_{ac} K_b^m \omega_m + K_{ab} h_c^g n^m \nabla_m \omega_g \end{aligned}$$

第四个等号利用了 $h_a^i \nabla_i h_k^g = n_k K_a^g + n^g K_{ik}$ 和 $h_a^b n_b = 0$ ，第五个等号运用了分部积分和 ω_a 是空间向量 $n^a \omega_a = 0$ （如果考虑类空的3+1的话，不同点会出在第四个等号展开里面部分的时候的正负号）

上面得到的结果，两边同时乘2并对指标ab取反对称得到

$$R^{(3)}_{abc}{}^d = h_a^e h_b^f h_c^l h_d^m R_{eflm} - 2K_{c[a} K_{b]}{}^d$$

其中利用了 $K_{[ab]} = 0$

把指标降下来就是

$$R^{(3)}_{abcd} = h_a^e h_b^f h_c^l h_d^m R_{eflm} - 2K_{c[a} K_{b]d}$$

于是我们得到了Gauss Relation

方程1.1

$$h_a^e h_b^f h_c^l h_d^m R_{eflm} = R^{(3)}_{abcd} + 2K_{c[a} K_{b]d}$$

Sec2 Codazzi Relation-1

往证

方程2.1

$$R_{\alpha\beta\gamma\delta}h^\alpha_\mu h^\beta_\nu h^\gamma_\rho n^\delta = -2D_{[\mu}K_{\nu]\rho}$$

定义一个辅助量

$$a_b = n^c \nabla_c n_b$$

Proff

$$\begin{aligned} R_{\alpha\beta\gamma\delta}h^\alpha_\mu h^\beta_\nu h^\gamma_\rho n^\delta &= h^\alpha_\mu h^\beta_\nu h^\gamma_\rho [\nabla_\alpha \nabla_\beta n_\gamma - \nabla_\beta \nabla_\alpha n_\gamma] \\ &= h^\alpha_\mu h^\beta_\nu h^\gamma_\rho [\nabla_\alpha (K_{\beta\gamma} - n_\beta a_\gamma) - \nabla_\beta (K_{\alpha\gamma} - n_\alpha a_\gamma)] \\ &= h^\alpha_\mu h^\beta_\nu h^\gamma_\rho (\nabla_\alpha K_{\beta\gamma} - \nabla_\beta K_{\alpha\gamma}) - a_\rho K_{\mu\nu} + a_\rho K_{\mu\nu} \\ &= D_\mu K_{\nu\rho} - D_\nu K_{\mu\rho} \\ &= 2D_{[\mu}K_{\nu]\rho} \end{aligned}$$

结果上差了一个负号，这个可以理解为使黎曼曲率的定义差一个负号，但是这样的话为什么前面的Gauss Relation 为什么不差负号呢？ 嘶，是因为他的外曲率和我定义差一个负号

$$K_{\mu\nu} = -\frac{1}{2}\mathcal{L}_n \gamma_{\mu\nu} = -\perp_\mu^\alpha \perp_\nu^\beta \nabla_\alpha n_\beta \quad (15)$$

证明一个上面用到的引理

引理2.1

$$\nabla_\beta n_\gamma = K_{\beta\gamma} - n_\beta a_\gamma$$

Proff

$$\begin{aligned} K_{\beta\gamma} &= h_\beta^\alpha \nabla_\alpha n_\gamma \\ &= (\delta_\beta^\alpha + n_\beta n^\alpha) \nabla_\alpha n_\gamma \\ &= \nabla_\beta n_\gamma + n_\beta n^\alpha \nabla_\alpha n_\gamma \\ &= \nabla_\beta n_\gamma + n_\beta a_\gamma \end{aligned}$$

证毕

Sec3.1 Codazzi Relation-2

引理3.1.1

$$h^\alpha_\mu h^\gamma_\rho n^\beta [\nabla_\alpha K_{\beta\gamma} - \nabla_\beta K_{\alpha\gamma}] = -\mathcal{L}_n K_{\mu\rho} + K_{\mu\alpha} K^\alpha_\rho$$

Proff

$$h^\alpha_\mu h^\gamma_\rho n^\beta [\nabla_\alpha K_{\beta\gamma} - \nabla_\beta K_{\alpha\gamma}] = h^\alpha_\mu h^\gamma_\rho n^\beta \nabla_\alpha K_{\beta\gamma} - h^\alpha_\mu h^\gamma_\rho n^\beta \nabla_\beta K_{\alpha\gamma}$$

其中第一部分

$$\begin{aligned} h^\alpha_\mu h^\gamma_\rho n^\beta \nabla_\alpha K_{\beta\gamma} &= -h^\alpha_\mu h^\gamma_\rho K_{\beta\gamma} \nabla_\alpha n^\beta \\ &= -h^\alpha_\mu K_{\beta\rho} \nabla_\alpha n^\beta \\ &= -K_\mu^\beta K_{\beta\rho} \end{aligned}$$

第二部分

$$\begin{aligned} h^\alpha_\mu h^\gamma_\rho n^\beta \nabla_\beta K_{\alpha\gamma} &= (\delta^\alpha_\mu + n^\alpha n_\mu) (\delta^\gamma_\rho + n^\gamma n_\rho) n^\beta \nabla_\beta K_{\alpha\gamma} \\ &= n^\beta \nabla_\beta K_{\mu\rho} + n^\alpha n^\beta n_\mu \nabla_\beta K_{\alpha\rho} + n^\gamma n^\beta n_\rho \nabla_\beta K_{\mu\gamma} + n^\alpha n^\beta n^\gamma n_\mu n_\rho \nabla_\beta K_{\alpha\gamma} \\ &= \mathcal{L}_n K_{\mu\rho} - K_{\beta\rho} \nabla_\mu n^\beta - K_{\mu\beta} \nabla_\rho n^\beta + n^\alpha n^\beta n_\mu \nabla_\beta K_{\alpha\rho} + n^\gamma n^\beta n_\rho \nabla_\beta K_{\mu\gamma} \end{aligned}$$

最后一项扔掉是因为，通过 $n^\alpha \nabla_\beta K_{\alpha\gamma} = \nabla_\beta (n^\alpha K_{\alpha\gamma}) - K_{\alpha\gamma} \nabla_\beta n^\alpha$ 不难看出 $n^\alpha n^\beta n^\gamma n_\mu n_\rho \nabla_\beta K_{\alpha\gamma} = 0$

倒数第二项

$$\begin{aligned} n^\alpha n^\beta n_\mu \nabla_\beta K_{\alpha\rho} &= n^\alpha (h^\beta_\mu - \delta^\beta_\mu) \nabla_\beta K_{\alpha\rho} \\ &= n^\alpha h^\beta_\mu \nabla_\beta K_{\alpha\rho} - n^\alpha \nabla_\mu K_{\alpha\rho} \\ &= -K_{\alpha\rho} h^\beta_\mu \nabla_\beta n^\alpha + K_{\alpha\rho} \nabla_\mu n^\alpha \\ &= -K_{\alpha\rho} K_\mu^\alpha + K_{\alpha\rho} \nabla_\mu n^\alpha \end{aligned}$$

同理可以得到倒数第一项满足

$$n^\gamma n^\beta n_\rho \nabla_\beta K_{\mu\gamma} = -K_{\mu\gamma} K_\rho^\gamma + K_{\mu\gamma} \nabla_\rho n^\gamma$$

由以上四式带入求和即得引理所需证明等式

$$h^\alpha_\mu h^\gamma_\rho n^\beta [\nabla_\alpha K_{\beta\gamma} - \nabla_\beta K_{\alpha\gamma}] = -\mathcal{L}_n K_{\mu\rho} + K_{\mu\alpha} K^\alpha_\rho$$

证毕

引理3.1.2

$$h^\alpha_\mu h^\gamma_\rho n^\beta [-\nabla_\alpha (n_\beta a_\gamma) + \nabla_\beta (n_\alpha a_\gamma)] = a_\mu a_\rho + h^\gamma_\rho h^\beta_\mu \nabla_\beta a_\gamma$$

Proff

$$\begin{aligned} & h^\alpha_\mu h^\gamma_\rho n^\beta [-\nabla_\alpha (n_\beta a_\gamma) + \nabla_\beta (n_\alpha a_\gamma)] \\ &= h^\gamma_\rho (\delta^\alpha_\mu + n^\alpha n_\mu) n^\beta [-\nabla_\alpha (n_\beta a_\gamma) + \nabla_\beta (n_\alpha a_\gamma)] \\ &= h^\gamma_\rho n^\beta [-\nabla_\mu (n_\beta a_\gamma) + \nabla_\beta (n_\mu a_\gamma)] \\ &= h^\gamma_\rho n^\beta [(\nabla_\beta n_\mu) a_\gamma + n_\mu \nabla_\beta a_\gamma - (\nabla_\mu n_\beta) a_\gamma - n_\beta \nabla_\mu a_\gamma] \\ &= h^\gamma_\rho a_\mu a_\gamma + h^\gamma_\rho n_\mu n^\beta \nabla_\beta a_\gamma - 0 + h^\gamma_\rho \nabla_\mu a_\gamma \\ &= a_\mu a_\rho + h^\gamma_\rho n_\mu n^\beta \nabla_\beta a_\gamma + h^\gamma_\rho \nabla_\mu a_\gamma \\ &= a_\mu a_\rho + h^\gamma_\rho (h_\mu^\beta - \delta_\mu^\beta) \nabla_\beta a_\gamma + h^\gamma_\rho \nabla_\mu a_\gamma \\ &= a_\mu a_\rho + h^\gamma_\rho h^\beta_\mu \nabla_\beta a_\gamma \\ &= a_\mu a_\rho + D_\mu a_\rho \end{aligned}$$

其中第二个等号是利用 α 和 β 的对称和反对称关系

引理3.1.3 (似乎并不成立)

$$a_\mu a_\rho + h^\gamma_\rho h^\beta_\mu \nabla_\beta a_\gamma = 0$$

也就是

$$a_\mu a_\rho + D_\mu a_\rho = 0, \quad a_b = n^c \nabla_c n_b$$

Proff

并不成立，我目前认为文章0810.1790中出现要使用这个结果是因为计算引理2.1的时候后面第二项被错误的认作是0了

Sec4 Contracted Codazzi Relation-1

from Gauss Relation

$$h_a^e h_b^f h_c^l h_d^m R_{eflm} = R^{(3)}_{abcd} + 2K_{c[a} K_{b]d}$$

indices b and d , we get

$$g^{bd} h_a^e h_b^f h_c^l h_d^m R_{eflm} = h_a^e h_c^l h^{fm} R_{eflm} = h_a^e h_c^l R_{el} + h_a^e h_c^l n^f n^m R_{eflm}$$

on the other side

$$g^{bd} (R^{(3)}_{abcd} + 2K_{c[a} K_{b]d}) = R^{(3)}_{ac} + K_{ac} K - K_{ab} K^b_c$$

write together, we get (4.1)

$$h_a^e h_c^l R_{el} + h_a^e h_c^l n^f n^m R_{eflm} = R^{(3)}_{ac} + K_{ac} K - K_{ab} K^b_c$$

use Codazzi Relation-2 can simplify this equations

we call it Contracted Codazzi Relation-1

Sec5 Contracted Codazzi Relation-2

from Codazzi relation 2.1

$$R_{\alpha\beta\gamma\delta}h^\alpha{}_\mu h^\beta{}_\nu h^\gamma{}_\rho n^\delta = -2D_{[\mu}K_{\nu]\rho}$$

we noticed, Contracting the Codazzi relation (2.1) on the indices μ and ρ yields to

$$\begin{aligned} g^{\mu\rho(4)}R_{\alpha\beta\gamma\delta}h^\alpha{}_\mu h^\beta{}_\nu h^\gamma{}_\rho n^\delta &= R_{\alpha\beta\gamma\delta}h^\beta{}_\nu h^{\alpha\gamma}n^\delta \\ &= R_{\beta\delta}h^\beta{}_\nu n^\delta + R_{\alpha\beta\gamma\delta}h^\beta{}_\nu n^\delta n^\alpha n^\gamma \\ &= R_{\beta\delta}h^\beta{}_\nu n^\delta \end{aligned}$$

on the other side

$$-2g^{\mu\rho}D_{[\mu}K_{\nu]\rho} = g^{\mu\rho}(D_\nu K_{\mu\rho} - D_\mu K_{\nu\rho}) = D_\nu K - D_\mu K^\mu{}_\nu$$

thus, we get

$$R_{\alpha\beta}h^\alpha{}_\nu n^\beta = D_\nu K - D_\mu K^\mu{}_\nu$$

we call it Contracted Codazzi Relation-2

Sec6 Riemann curvature scalar

To get Riemann curvature, we use g^{ac} indices (4.1)

$$h_a{}^e h_c{}^l R_{el} + h_a{}^e h_c{}^l n^f n^m R_{eflm} = R^{(3)}_{ac} + K_{ac}K - K_{ab}K^b{}_c$$

on the left side, we get

$$g^{ac}(h_a{}^e h_c{}^l R_{el} + h_a{}^e h_c{}^l n^f n^m R_{eflm}) = {}^{(4)}R + 2n^e n^l R_{el}$$

on the right side, we get

$$R^{(3)}_{ac} + K_{ac}K - K_{ab}K^b{}_c \rightarrow {}^{(3)}R + K^2 - K_{ab}K^{ab}$$

that means (6.1)

$${}^{(4)}R = {}^{(3)}R + K^2 - K_{ab}K^{ab} - 2n^e n^l R_{el}$$

where $n^e n^l R_{el}$ contains $K^2 - K_{ab}K^{ab}$ and a boundary term

由（这部分边界项的证明可以看梁书）

$$n^a R_{ab}n^b = -\nabla_a(n^a \nabla_c n^c) + K^2 + \nabla_c(n^a \nabla_a n^c) - K_{ac}K^{ac}$$

可以得到

$$\mathcal{L}_{GR} = \sqrt{h}N(R^{(3)} + K_{ab}K^{ab} - K^2) + 2\sqrt{-g}[\nabla_a(n^a \nabla_c n^c) - \nabla_c(n^a \nabla_a n^c)]$$

计算过程可以看[3+1分解和拉氏形式](#)，我在这个的开篇有记下来这段的计算

把前面计算的结果在这里也放一份

$$R_{abcd}h^a{}_e h^c{}_f n^b n^d = \nabla_e(n^b \nabla_b n_f) - (\nabla_e n^b)(\nabla_b n_f) - \nabla_b(n^b \nabla_e n_f) + (\nabla_b n^b)(\nabla_e n_f)$$

Sec7 R^2 Term

consider (6.1)

$${}^{(4)}R = {}^{(3)}R + K^2 - K_{ab}K^{ab} - 2n^a n^c R_{ac}$$

in which

$$\begin{aligned} n^a n^c R_{ac} &= n^a n^c g^{bd} R_{abcd} \\ &= n^a n^c (h^{bd} + n^b n^d) R_{abcd} \\ &= n^a n^c h^{bd} R_{abcd} \\ &= h^{ef} n^a n^c h^b{}_e h^d{}_f R_{abcd} \\ &\equiv F_0 \end{aligned}$$

于是

$$\begin{aligned}
{}^{(4)}R^2 &= {}^{(3)}R^2 + K^4 + (K_{ab}K^{ab})^2 + 4(F_0)^2 \\
&\quad + 2{}^{(3)}RK^2 - 2{}^{(3)}RK_{ab}K^{ab} - 4{}^{(3)}RF_0 \\
&\quad - 2K^2K_{ab}K^{ab} - 4K^2F_0 + 4K_{ab}K^{ab}F_0
\end{aligned}$$

这里的关键是计算出 F_0 的表达式，其实和后面的都是一样的需求

$$F_0 = K^2 - K_{ac}K^{ac} - \nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)$$

于是（省略掉 ${}^{(3)}$ ）

$$\begin{aligned}
{}^{(4)}R^2 &= {}^{(3)}R^2 + K^4 + (K_{ab}K^{ab})^2 \\
&\quad + 4[K^2 - K_{ac}K^{ac} - \nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)]^2 \\
&\quad + 2{}^{(3)}RK^2 - 2{}^{(3)}RK_{ab}K^{ab} - 4{}^{(3)}R[K^2 - K_{ac}K^{ac} - \nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)] \\
&\quad - 2K^2K_{ab}K^{ab} - 4K^2[K^2 - K_{ac}K^{ac} - \nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)] \\
&\quad + 4K_{ab}K^{ab}[K^2 - K_{ac}K^{ac} - \nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)] \\
&= R^2 + K^4 + (K_{ab}K^{ab})^2 \\
&\quad + 4[K^2 - K_{ac}K^{ac} - \nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)]^2 \\
&\quad + 2RK^2 - 2RK_{ab}K^{ab} - 4R[K^2 - K_{ac}K^{ac} - \nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)] \\
&\quad - 2K^2K_{ab}K^{ab} - 4K^2[K^2 - K_{ac}K^{ac} - \nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)] \\
&\quad + 4K_{ab}K^{ab}[K^2 - K_{ac}K^{ac} - \nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)]
\end{aligned}$$

这里有一个简单一点的算法，记

$$F_{01} = -\nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)$$

有

$${}^{(4)}R = {}^{(3)}R - K^2 + K_{ab}K^{ab} - 2F_{01}$$

也就意味着

$$\begin{aligned}
{}^{(4)}R^2 &= R^2 + K^4 + (K_{ab}K^{ab})^2 + 4(F_{01})^2 \\
&\quad - 2RK^2 + 2RK_{ab}K^{ab} - 4RF_{01} \\
&\quad - 2K^2K_{ab}K^{ab} + 4K^2F_{01} - 4K_{ab}K^{ab}F_{01}
\end{aligned}$$

再带入 F_{01} 的形式

$$\begin{aligned}
{}^{(4)}R^2 &= R^2 + K^4 + (K_{ab}K^{ab})^2 + 4[-\nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)]^2 \\
&\quad - 2RK^2 + 2RK_{ab}K^{ab} - 4R[-\nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)] \\
&\quad - 2K^2K_{ab}K^{ab} + 4K^2[-\nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)] \\
&\quad - 4K_{ab}K^{ab}[-\nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)]
\end{aligned}$$

这里面并没有很容易看出来可以直接消掉的项，先去计算一下后面的试试看

Sec8 $R_{ab}R^{ab}$ Term

consider

$$\begin{aligned}
R_{ab}R^{ab} &= g^{ac}g^{bd}R_{ab}R_{cd} = g^{ac}g^{bd}g^{ef}g^{ij}R_{aebf}R_{cidj} \\
&= (h^{ac} + n^an^c)(h^{bd} + n^bn^d)(h^{ef} + n^en^f)(h^{ij} + n^in^j)R_{aebf}R_{cidj} \\
&= h^{ac}h^{bd}h^{ef}h^{ij}R_{aebf}R_{cidj} \\
&\quad + n^an^ch^{bd}h^{ef}h^{ij}R_{aebf}R_{cidj} + n^bn^dh^{ac}h^{ef}h^{ij}R_{aebf}R_{cidj} \\
&\quad + n^en^fh^{ac}h^{bd}h^{ij}R_{aebf}R_{cidj} + n^in^jh^{ac}h^{bd}h^{ef}R_{aebf}R_{cidj} \\
&\quad + n^an^bn^cn^dh^{ef}h^{ij}R_{aebf}R_{cidj} + n^en^fn^in^jh^{ac}h^{bd}R_{aebf}R_{cidj} \\
&= h^{ac}h^{bd}h^{ef}h^{ij}R_{aebf}R_{cidj} \\
&\quad + 2n^an^ch^{bd}h^{ef}h^{ij}R_{aebf}R_{cidj} + 2n^en^fn^{ac}h^{bd}h^{ij}R_{aebf}R_{cidj} \\
&\quad + n^an^bn^cn^dh^{ef}h^{ij}R_{aebf}R_{cidj} + n^en^fn^in^jh^{ac}h^{bd}R_{aebf}R_{cidj} \\
&= S_1 + 2S_2 + 2S_3 + S_4 + S_5
\end{aligned}$$

分别计算这五项，第一项

$$\begin{aligned}
S_1 &= h^{ac} h^{bd} h^{ef} h^{ij} R_{aebf} R_{cidj} \\
&= h^{kl} h^{mn} h^{op} h^{qr} h_k^a h_l^c h_m^b h_n^d h_o^e h_p^f h_q^i h_r^j R_{aebf} R_{cidj} \\
&= h^{kl} h^{mn} h^{op} h^{qr} (h_k^a h_o^e h_m^b h_p^f R_{aebf}) (h_l^c h_q^i h_n^d h_r^j R_{cidj}) \\
&= h^{kl} h^{mn} h^{op} h^{qr} (R^{(3)}_{komp} + 2K_{m[k} K_{o]p}) (R^{(3)}_{lqnr} + 2K_{n[l} K_{q]r}) \\
&= R^{(3)}_{ab} R^{(3)ab} + 2R_{ab} (K^{ab} K - K^{bc} K^a_c) + (K_{ab} K - K_{ac} K_b^c) (K^{ab} K - K^{ad} K^b_d)
\end{aligned}$$

第二项

$$\begin{aligned}
S_2 &= n^a n^c h^{bd} h^{ef} h^{ij} R_{aebf} R_{cidj} \\
&= n^a n^c h^{kl} h^{mn} h^{op} h_k^b h_l^d h_m^e h_n^f h_o^i h_p^j R_{aebf} R_{cidj} \\
&= h^{kl} h^{mn} h^{op} (n^a h_m^e h_k^b h_n^f R_{aebf}) (n^c h_o^i h_l^d h_p^j R_{cidj}) \\
&= g^{kl} g^{mn} g^{op} (2D_{[k} K_{n]m}) (2D_{[l} K_{p]o})
\end{aligned}$$

第三项

$$\begin{aligned}
S_3 &= n^e n^f h^{ac} h^{bd} h^{ij} R_{aebf} R_{cidj} \\
&= n^e n^f h^{kl} h^{mn} h^{op} h_k^a h_l^c h_m^b h_n^d h_o^i h_p^j R_{aebf} R_{cidj} \\
&= h^{kl} h^{mn} h^{op} (n^e n^f h_k^a h_m^b R_{aebf}) (h_l^c h_o^i h_n^d h_p^j R_{cidj})
\end{aligned}$$

注意

$$\begin{aligned}
R_{abcd} h^a_e h^c_f h^b_n n^d &= \nabla_e (n^b \nabla_b n_f) - (\nabla_e n^b) (\nabla_b n_f) - \nabla_b (n^b \nabla_e n_f) + (\nabla_b n^b) (\nabla_e n_f) \\
R_{aebf} h^a_k h^b_m n^e n^f &= \nabla_k (n^e \nabla_e n_m) - (\nabla_k n^e) (\nabla_e n_m) - \nabla_e (n^e \nabla_k n_m) + (\nabla_e n^e) (\nabla_k n_m)
\end{aligned}$$

和

$$\begin{aligned}
h^e_a h^f_b h^i_c h^d_m R_{eflm} &= R^{(3)}_{abcd} + 2K_{c[a} K_{b]d} \\
h^i_l h^c_o h^d_n h^j_p R_{cidj} &= R^{(3)}_{lonp} + 2K_{n[l} K_{o]p}
\end{aligned}$$

带回去

$$\begin{aligned}
S_3 &= h^{kl} h^{mn} h^{op} (n^e n^f h_k^a h_m^b R_{aebf}) (h_l^c h_o^i h_n^d h_p^j R_{cidj}) \\
&= g^{kl} g^{mn} g^{op} [\nabla_k (n^e \nabla_e n_m) - (\nabla_k n^e) (\nabla_e n_m) - \nabla_e (n^e \nabla_k n_m) + (\nabla_e n^e) (\nabla_k n_m)] \\
&\quad [R^{(3)}_{lonp} + K_{nl} K_{op} - K_{no} K_{lp}]
\end{aligned}$$

其中第二个等号是因为后面的h可以和前面需要补的n正交

第四项

$$\begin{aligned}
S_4 &= n^a n^b n^c n^d h^{ef} h^{ij} R_{aebf} R_{cidj} \\
&= n^a n^b n^c n^d h^{kl} h^{mn} h_k^e h_l^f h_m^i h_n^j R_{aebf} R_{cidj} \\
&= h^{kl} h^{mn} (n^a n^b h_k^e h_l^f R_{aebf}) (n^c n^d h_m^i h_n^j R_{cidj})
\end{aligned}$$

其中

$$\begin{aligned}
R_{aebf} h^a_k h^b_m n^e n^f &= \nabla_k (n^e \nabla_e n_m) - (\nabla_k n^e) (\nabla_e n_m) - \nabla_e (n^e \nabla_k n_m) + (\nabla_e n^e) (\nabla_k n_m) \\
R_{aebf} h^e_k h^f_m n^a n^b &= \nabla_k (n^e \nabla_e n_l) - (\nabla_k n^e) (\nabla_e n_l) - \nabla_e (n^e \nabla_k n_l) + (\nabla_e n^e) (\nabla_k n_l) \\
R_{cidj} h^i_m h^j_n n^c n^d &= \nabla_m (n^o \nabla_o n_n) - (\nabla_m n^o) (\nabla_o n_n) - \nabla_o (n^o \nabla_m n_n) + (\nabla_o n^o) (\nabla_m n_n)
\end{aligned}$$

带回去就得到

$$S_4 = [K^2 - K_{ac} K^{ac} - \nabla_a (n^a \nabla_c n^c) + \nabla_c (n^a \nabla_a n^c)]^2$$

第五项

$$\begin{aligned}
S_5 &= n^e n^f n^i n^j h^{ac} h^{bd} R_{aebf} R_{cidj} \\
&= n^e n^f n^i n^j h^{kl} h^{mn} h_k^a h_l^c h_m^b h_n^d R_{aebf} R_{cidj} \\
&= h^{kl} h^{mn} (n^e n^f h_k^a h_m^b R_{aebf}) (n^i n^j h_l^c h_n^d R_{cidj})
\end{aligned}$$

注意到

$$\begin{aligned}
R_{aebf} h^a_k h^b_m n^e n^f &= \nabla_k (n^e \nabla_e n_m) - (\nabla_k n^e) (\nabla_e n_m) - \nabla_e (n^e \nabla_k n_m) + (\nabla_e n^e) (\nabla_k n_m) \\
R_{cidj} h^c_l h^d_n n^i n^j &= \nabla_l (n^o \nabla_o n_n) - (\nabla_l n^o) (\nabla_o n_n) - \nabla_o (n^o \nabla_l n_n) + (\nabla_o n^o) (\nabla_l n_n)
\end{aligned}$$

带回去得到

$$S_5 = g^{kl}g^{mn}[\nabla_k(n^e\nabla_en_m) - (\nabla_kn^e)(\nabla_en_m) - \nabla_e(n^e\nabla_kn_m) + (\nabla_en^e)(\nabla_kn_m)] \\ [\nabla_l(n^o\nabla_on_n) - (\nabla_ln^o)(\nabla_on_n) - \nabla_o(n^o\nabla_ln_n) + (\nabla_on^o)(\nabla_ln_n)]$$

综合上面的计算结果，我们可以写出

$$\begin{aligned} & {}^{(4)}R_{ab}R^{ab} \\ &= S_1 + 2S_2 + 2S_3 + S_4 + S_5 \\ &= R_{ab}R^{ab} + 2R_{ab}(K^{ab}K - K^{bc}K^a{}_c) + (K_{ab}K - K_{ac}K^c{}_b)(K^{ab}K - K^{ad}K^b{}_d) \\ &\quad + 2g^{kl}g^{mn}g^{op}(2D_{[k}K_{n]m})(2D_{[l}K_{p]o}) \\ &\quad + 2g^{kl}g^{mn}g^{op}[\nabla_k(n^e\nabla_en_m) - (\nabla_kn^e)(\nabla_en_m) - \nabla_e(n^e\nabla_kn_m) + (\nabla_en^e)(\nabla_kn_m)] \\ &\quad [R^{(3)}_{lonp} + K_{nl}K_{op} - K_{no}K_{lp}] \\ &\quad + [K^2 - K_{ac}K^{ac} - \nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)]^2 \\ &\quad + g^{kl}g^{mn}[\nabla_k(n^e\nabla_en_m) - (\nabla_kn^e)(\nabla_en_m) - \nabla_e(n^e\nabla_kn_m) + (\nabla_en^e)(\nabla_kn_m)] \\ &\quad [\nabla_l(n^o\nabla_on_n) - (\nabla_ln^o)(\nabla_on_n) - \nabla_o(n^o\nabla_ln_n) + (\nabla_on^o)(\nabla_ln_n)] \end{aligned}$$

Sec9 $R_{abcd}R^{abcd}$ Term

noticed (左边写 h^{ab} 也行)

$$\begin{aligned} g^{ab}h_a{}^ch_b{}^e &= g^{ce} + g^{cb}n_bn^e + g^{ae}n_an^c + g^{ab}n_an_bn^cn^e \\ &= g^{ce} + n^cn^e \\ &= h^{ce} \end{aligned}$$

consider

$$\begin{aligned} R_{abcd}R^{abcd} &= g^{ae}g^{bf}g^{cg}g^{dh}R_{abcd}R_{efgh} \\ &= (h^{ae} + n^an^e)(h^{bf} + n^bn^f)(h^{cg} + n^cn^g)(h^{dh} + n^dn^h)R_{abcd}R_{efgh} \\ &= h^{ae}h^{bf}h^{cg}h^{dh}R_{abcd}R_{efgh} \\ &\quad + n^an^eh^{bf}h^{cg}h^{dh}R_{abcd}R_{efgh} + n^bn^fh^{ae}h^{cg}h^{dh}R_{abcd}R_{efgh} \\ &\quad + n^cn^gh^{ae}h^{bf}h^{dh}R_{abcd}R_{efgh} + n^dn^h^{ae}h^{bf}h^{cg}R_{abcd}R_{efgh} \\ &\quad + 2n^an^en^cn^gh^{bf}h^{dh}R_{abcd}R_{efgh} + 2n^bn^fn^cn^gh^{ae}h^{dh}R_{abcd}R_{efgh} \\ &= h^{ae}h^{bf}h^{cg}h^{dh}R_{abcd}R_{efgh} \\ &\quad + 4n^an^eh^{bf}h^{cg}h^{dh}R_{abcd}R_{efgh} + 4n^an^en^cn^gh^{bf}h^{dh}R_{abcd}R_{efgh} \\ &\equiv T_1 + 4T_2 + 4T_3 \end{aligned}$$

分成三项计算，第一项 use Gauss relation

$$\begin{aligned} T_1 &= h^{ae}h^{bf}h^{cg}h^{dh}R_{abcd}R_{efgh} \\ &= g^{ij}g^{kl}g^{mn}g^{op}h_i{}^ah_j{}^eh_k{}^bh_l{}^fh_m{}^ch_n{}^gh_o{}^dh_p{}^hR_{abcd}R_{efgh} \\ &= g^{ij}g^{kl}g^{mn}g^{op}({}^{(3)}R_{ikmo} + 2K_{m[i}K_{k]o})({}^{(3)}R_{jlnp} + 2K_{n[j}K_{l]p}) \\ &= h^{ij}h^{kl}h^{mn}h^{op}({}^{(3)}R_{ikmo} + 2K_{m[i}K_{k]o})({}^{(3)}R_{jlnp} + 2K_{n[j}K_{l]p}) \\ &= ({}^{(3)}R_{abcd}{}^{(3)}R^{abcd} + 4({}^{(3)}R_{abcd}K^c{}^{[a}K^{b]d} + 4(K_{a[b}K_{c]d})(K^{a[b}K^{c]d})) \end{aligned}$$

第二项 use Codazzi relation

$$\begin{aligned} T_2 &= n^an^eh^{bf}h^{cg}h^{dh}R_{abcd}R_{efgh} \\ &= n^an^eh^{ij}h^{kl}h^{mn}h_i{}^bh_j{}^fh_k{}^ch_l{}^gh_m{}^dh_n{}^hR_{abcd}R_{efgh} \\ &= g^{ij}g^{kl}g^{mn}(n^ah_i{}^bh_k{}^ch_m{}^dR_{abcd})(n^eh_j{}^fh_l{}^gh_n{}^hR_{efgh}) \\ &= g^{ij}g^{kl}g^{mn}(2D_{[k}K_{m]i})(2D_{[l}K_{n]j}) \end{aligned}$$

第三项

$$\begin{aligned} T_3 &= n^an^en^cn^gh^{bf}h^{dh}R_{abcd}R_{efgh} \\ &= h^{ij}h^{kl}(n^an^ch_i{}^bh_k{}^dR_{abcd})(n^en^gh_j{}^fh_l{}^hR_{efgh}) \\ &= S_5 \\ &= g^{kl}g^{mn}[\nabla_k(n^e\nabla_en_m) - (\nabla_kn^e)(\nabla_en_m) - \nabla_e(n^e\nabla_kn_m) + (\nabla_en^e)(\nabla_kn_m)] \\ &\quad [\nabla_l(n^o\nabla_on_n) - (\nabla_ln^o)(\nabla_on_n) - \nabla_o(n^o\nabla_ln_n) + (\nabla_on^o)(\nabla_ln_n)] \end{aligned}$$

综合前面算的可以写出4

$$\begin{aligned}
& {}^{(4)}R_{abcd}R^{abcd} \\
&= T_1 + 4T_2 + 4T_3 \\
&= {}^{(3)}R_{abcd}{}^{(3)}R^{abcd} + 4{}^{(3)}R_{abcd}K^c[aK^b]d + 4(K_{a[b}K_{c]d})(K^{a[b}K^{c]d}) \\
&\quad + g^{ij}g^{kl}g^{mn}(2D_{[k}K_{m]i})(2D_{[l}K_{n]j}) \\
&\quad + 4g^{kl}g^{mn}[\nabla_k(n^e\nabla_en_m) - (\nabla_kn^e)(\nabla_en_m) - \nabla_e(n^e\nabla_kn_m) + (\nabla_en^e)(\nabla_kn_m)] \\
&\quad [\nabla_l(n^o\nabla_on_n) - (\nabla_ln^o)(\nabla_on_n) - \nabla_o(n^o\nabla_ln_n) + (\nabla_on^o)(\nabla_ln_n)]
\end{aligned}$$

Gauss-Bonnet Term

先写下之前算好的这些项

标量曲率平方

$$\begin{aligned}
{}^{(4)}R^2 &= R^2 + K^4 + (K_{ab}K^{ab})^2 + 4[-\nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)]^2 \\
&\quad - 2RK^2 + 2RK_{ab}K^{ab} - 4R[-\nabla_a(n^a\nabla_en^c) + \nabla_c(n^a\nabla_an^c)] \\
&\quad - 2K^2K_{ab}K^{ab} + 4K^2[-\nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)] \\
&\quad - 4K_{ab}K^{ab}[-\nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)]
\end{aligned}$$

里奇张量缩并

$$\begin{aligned}
& {}^{(4)}R_{ab}R^{ab} \\
&= R_{ab}R^{ab} + 2R_{ab}(K^{ab}K - K^{bc}K^a{}_c) + (K_{ab}K - K_{ac}K_b^c)(K^{ab}K - K^{ad}K^b{}_d) \\
&\quad + 2g^{kl}g^{mn}g^{op}(2D_{[k}K_{n]m})(2D_{[l}K_{p]o}) \\
&\quad + 2g^{kl}g^{mn}g^{op}[\nabla_k(n^e\nabla_en_m) - (\nabla_kn^e)(\nabla_en_m) - \nabla_e(n^e\nabla_kn_m) + (\nabla_en^e)(\nabla_kn_m)] \\
&\quad [R^{(3)}_{lonp} + K_{nl}K_{op} - K_{no}K_{lp}] \\
&\quad + [K^2 - K_{ac}K^{ac} - \nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)]^2 \\
&\quad + g^{kl}g^{mn}[\nabla_k(n^e\nabla_en_m) - (\nabla_kn^e)(\nabla_en_m) - \nabla_e(n^e\nabla_kn_m) + (\nabla_en^e)(\nabla_kn_m)] \\
&\quad [\nabla_l(n^o\nabla_on_n) - (\nabla_ln^o)(\nabla_on_n) - \nabla_o(n^o\nabla_ln_n) + (\nabla_on^o)(\nabla_ln_n)]
\end{aligned}$$

黎曼曲率缩并

$$\begin{aligned}
& {}^{(4)}R_{abcd}R^{abcd} \\
&= {}^{(3)}R_{abcd}{}^{(3)}R^{abcd} + 4{}^{(3)}R_{abcd}K^c[aK^b]d + 4(K_{a[b}K_{c]d})(K^{a[b}K^{c]d}) \\
&\quad + g^{ij}g^{kl}g^{mn}(2D_{[k}K_{m]i})(2D_{[l}K_{n]j}) \\
&\quad + 4g^{kl}g^{mn}[\nabla_k(n^e\nabla_en_m) - (\nabla_kn^e)(\nabla_en_m) - \nabla_e(n^e\nabla_kn_m) + (\nabla_en^e)(\nabla_kn_m)] \\
&\quad [\nabla_l(n^o\nabla_on_n) - (\nabla_ln^o)(\nabla_on_n) - \nabla_o(n^o\nabla_ln_n) + (\nabla_on^o)(\nabla_ln_n)]
\end{aligned}$$

那么Gauss-Bonnet项就是（后面没标括号的的就是空间曲率）

$$\begin{aligned}
\mathcal{L}_{GB} &= {}^{(4)}R^2 - 4{}^{(4)}R_{ab}R^{ab} + {}^{(4)}R_{abcd}R^{abcd} \\
&= R^2 + K^4 + (K_{ab}K^{ab})^2 + 4[-\nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)]^2 \\
&\quad - 2RK^2 + 2RK_{ab}K^{ab} - 4R[-\nabla_a(n^a\nabla_en^c) + \nabla_c(n^a\nabla_an^c)] \\
&\quad - 2K^2K_{ab}K^{ab} + 4K^2[-\nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)] \\
&\quad - 4K_{ab}K^{ab}[-\nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)] \\
&\quad - 4\{R_{ab}R^{ab} + 2R_{ab}(K^{ab}K - K^{bc}K^a{}_c) + (K_{ab}K - K_{ac}K_b^c)(K^{ab}K - K^{ad}K^b{}_d) \\
&\quad + 2g^{kl}g^{mn}g^{op}(2D_{[k}K_{n]m})(2D_{[l}K_{p]o}) \\
&\quad + 2g^{kl}g^{mn}g^{op}[\nabla_k(n^e\nabla_en_m) - (\nabla_kn^e)(\nabla_en_m) - \nabla_e(n^e\nabla_kn_m) + (\nabla_en^e)(\nabla_kn_m)] \\
&\quad [R_{lonp} + K_{nl}K_{op} - K_{no}K_{lp}] \\
&\quad + [K^2 - K_{ac}K^{ac} - \nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)]^2 \\
&\quad + g^{kl}g^{mn}[\nabla_k(n^e\nabla_en_m) - (\nabla_kn^e)(\nabla_en_m) - \nabla_e(n^e\nabla_kn_m) + (\nabla_en^e)(\nabla_kn_m)] \\
&\quad [\nabla_l(n^o\nabla_on_n) - (\nabla_ln^o)(\nabla_on_n) - \nabla_o(n^o\nabla_ln_n) + (\nabla_on^o)(\nabla_ln_n)]\} \\
&\quad + R_{abcd}R^{abcd} + 4R_{abcd}K^c[aK^b]d + 4(K_{a[b}K_{c]d})(K^{a[b}K^{c]d}) \\
&\quad + g^{ij}g^{kl}g^{mn}(2D_{[k}K_{m]i})(2D_{[l}K_{n]j}) \\
&\quad + 4g^{kl}g^{mn}[\nabla_k(n^e\nabla_en_m) - (\nabla_kn^e)(\nabla_en_m) - \nabla_e(n^e\nabla_kn_m) + (\nabla_en^e)(\nabla_kn_m)] \\
&\quad [\nabla_l(n^o\nabla_on_n) - (\nabla_ln^o)(\nabla_on_n) - \nabla_o(n^o\nabla_ln_n) + (\nabla_on^o)(\nabla_ln_n)]
\end{aligned}$$

整理

$$\begin{aligned}
\mathcal{L}_{GB} &= {}^{(4)}R^2 - 4{}^{(4)}R_{ab}R^{ab} + {}^{(4)}R_{abcd}R^{abcd} \\
&= R^2 + K^4 + (K_{ab}K^{ab})^2 + 4[-\nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)]^2 \\
&\quad - 2RK^2 + 2RK_{ab}K^{ab} - 4R[-\nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)] \\
&\quad - 2K^2K_{ab}K^{ab} + 4K^2[-\nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)] \\
&\quad - 4K_{ab}K^{ab}[-\nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)] \\
&\quad - 4\{R_{ab}R^{ab} + 2R_{ab}(K^{ab}K - K^{bc}K^a{}_c) + (K_{ab}K - K_{ac}K^b{}_c)(K^{ab}K - K^{ad}K^b{}_d) \\
&\quad + 2g^{kl}g^{mn}g^{op}(2D_{[k}K_{n]m})(2D_{[l}K_{p]o}) \\
&\quad + 2g^{kl}g^{mn}g^{op}[\nabla_k(n^e\nabla_en_m) - (\nabla_kn^e)(\nabla_en_m) - \nabla_e(n^e\nabla_kn_m) + (\nabla_en^e)(\nabla_kn_m)] \\
&\quad [R_{lonp} + K_{nl}K_{op} - K_{no}K_{lp}] \\
&\quad + [K^2 - K_{ac}K^{ac} - \nabla_a(n^a\nabla_cn^c) + \nabla_c(n^a\nabla_an^c)]^2\} \\
&\quad + R_{abcd}R^{abcd} + 4R_{abcd}K^{c[a}K^{b]d} + 4(K_{a[b}K_{c]d})(K^{a[b}K^{c]d}) \\
&\quad + g^{ij}g^{kl}g^{mn}(2D_{[k}K_{m]i})(2D_{[l}K_{n]j})
\end{aligned}$$

很抱歉还没有整理出来GB的3+1形式